

Accelerating PolyFEM Triangular Mesh Impact Simulation with Machine Learning
Surrogates

by

Jiacheng Zhao

B.Sc. (Computer Science and Technology), Beijing Institute of Technology, 2024

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We acknowledge and respect the Lək'wəḡən (Songhees and X^wsepsəm/Esquimalt)
Peoples on whose territory the university stands, and the Lək'wəḡən and W̱SÁNEĆ
Peoples whose historical relationships with the land continue to this day.

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ABSTRACT

Nonlinear transient finite element method (FEM) simulation of impact on triangular meshes is computationally expensive. In PolyFEM,²² a single forward solve takes on the order of minutes per design, and a combined forward and adjoint pass takes longer. This makes parameter sweeps and gradient based optimization, which each issue thousands to millions of evaluations, impractical to run on PolyFEM directly. This thesis develops two machine learning surrogates over the design parameters (θ, h, E) , namely mesh apex angle, wall thickness, and Young’s modulus. The surrogates replace each FEM call with sub-millisecond inference.

The *forward* surrogate predicts the peak Von Mises stress on the impacting block. XGBoost with twenty-two features informed by physics and a $\log(1 + y)$ target transform achieves $R^2 = 0.97$ and $\text{Acc}@10\% = 74.1\%$ on a test set that the model never saw during training. The *gradient* surrogate predicts the three adjoint sensitivities dL/dE , dL/dh , $dL/d\theta$ by decomposing each target into its sign and the logarithm of its magnitude, which handles gradients spanning six to nine orders of magnitude. The decomposition works regardless of model family. XGBoost and a multihead neural network agree within a few percentage points across the three components, with $\text{Acc}@20\%$ ranging from about 47% in the hardest regime past the bifurcation to about 75% on the smoother gradients.

Both surrogates show the same pattern. Difficulty tracks the underlying physics, and phase transitions impose a noise floor that no architecture can overcome. The thesis pairs the surrogates with a deployment analysis that matches each accuracy

regime to the workflows it supports, so that surrogates and PolyFEM are used together rather than as substitute.¹

¹Code, trained checkpoints, and reproduction instructions are available at <https://github.com/jczhao1/polyfem-ml-surrogate>.

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Compute Canada, for providing the computational resources that made the 30,000-plus FEM simulations underlying this work feasible.

My family and friends, for support, encouragement, and patience throughout the program.

DEDICATION

To my family.

Chapter 1

Introduction

Triangular mesh structures are widely adopted in packaging, protective equipment, and energy absorbing applications because of their tunable ratio of strength to weight and bending dominated impact response.^{6,8} Designing such structures for a target impact specification requires nonlinear transient finite element method (FEM) simulation with hyperelasticity at large deformation²¹ and frictional contact. Using PolyFEM,²² a single forward solve takes on the order of minutes per design, and a combined forward and adjoint solve that yields exact parameter sensitivities takes meaningfully longer. Paying this cost once per design point is acceptable, but most downstream tasks need many evaluations of the same simulation. A parameter sweep over (θ, h, E) , namely the mesh apex angle, wall thickness, and Young’s modulus, evaluates tens of thousands of design points; gradient based optimization and Bayesian search each issue thousands to millions of evaluations. Running these workloads on PolyFEM directly is several orders of magnitude too slow to be practical.

Machine learning surrogates can replace each FEM evaluation with millisecond inference and have been used effectively for forward prediction in lattice and mesh mechanics.^{2,9} Supporting a full optimization workflow requires predicting two distinct quantities. The *forward surrogate* predicts the peak Von Mises stress on the impacting block (body 2),

$$f_{\text{fwd}}: (\theta, h, E) \mapsto \sigma_{\text{peak}}^{\text{block}} = \max_{t \in [0, T]} \sigma_{\text{VM}}^{\text{body } 2}(t), \quad (1.1)$$

where $T = 0.5$ s is the simulated impact horizon. This quantity governs the transmitted force, which is the primary failure criterion for the protected payload in packaging. The *gradient surrogate* predicts the parameter sensitivities of a smooth temporal

maximum of the Von Mises objective on the mesh (body 1),

$$f_{\text{grad}}: (\theta, h, E) \mapsto \nabla_{(E, h, \theta)} L, \quad L = \text{smooth_max}_t \sigma_{\text{VM}}^{\text{body } 1}(t), \quad (1.2)$$

yielding the three components $(dL/dE, dL/dh, dL/d\theta)$. These are the inputs a gradient based optimizer consumes to update designs.

The two surrogates target the same parameter space but face distinct learning challenges. The forward target is a noisy nonmonotonic scalar shaped by a bifurcation between buckling modes near $\theta \approx 88^\circ$.⁸ The gradient targets span six to nine orders of magnitude and exhibit sign discontinuities, which a single MSE loss cannot resolve. Both effects are real features of the impact physics, not artifacts of the FEM solver that could be removed by refining the simulation, and a central thread of this thesis is that they impose ceilings that no model architecture can bypass.

This is where the thesis is positioned. Machine learning surrogates for FEM are by now well established, but most of that work targets smooth mappings from design parameters to response, where a high aggregate R^2 is easy to reach and reasonable to trust.^{9,15} The contribution here is not that a surrogate can replace FEM, which is known, but that it can be built to work across a structural discontinuity: a buckling bifurcation that makes the forward response a noisy valley, and gradients that flip sign and span many orders of magnitude across that same transition. Handling this discontinuous regime, rather than the smooth regime the surrogate literature usually assumes, is what shapes every methodological choice that follows, from the features informed by physics to the sign and log-magnitude decomposition to the deployment relevant metrics used in place of aggregate R^2 .

This thesis makes the following contributions, all supported empirically on test sets the model never saw during training, together with ablation studies that remove one component at a time.

- **Forward surrogate driven by engineering informed by physics.** An XGBoost⁴ model with twenty-two features motivated by physics and a $\log(1+y)$ target transform achieves $\text{Acc}@10\% = 74.1\%$ on 1,889 test samples the model never saw during training, +15.9 percentage points over the same model trained on the three raw inputs. Predicting the peak directly outperforms predicting the full 51-step transient and then extracting its peak, where a DeepONet¹⁷ reaches $R^2 = 0.92$ on the trajectory but only 39.5% $\text{Acc}@10\%$ on the extracted peak.

- **Gradient surrogate via sign and log-magnitude decomposition.** Decomposing each gradient target into its sign and the logarithm of its magnitude makes targets spanning six to nine orders of magnitude learnable, and the same decomposition works regardless of model family. XGBoost reaches within five percentage points of a multihead neural network on the hardest gradient ($dL/d\theta$).
- **A view of surrogate difficulty grounded in the physics.** Smooth regions yield nearly perfect prediction ($\text{Acc@10\%} \approx 99\%$ for $\theta < 80^\circ$), whereas regions near the buckling bifurcation are irreducibly difficult ($\text{Acc@10\%} = 59.6\%$ on 458 test samples in $[87^\circ, 89^\circ)$). Both XGBoost and the multihead neural network show large drops on the $dL/d\theta$ component past the bifurcation, falling far below the ceilings reached on the smoother gradients, which is consistent with a difficulty imposed by the physics rather than a limitation of model capacity.

Chapter 2

Problem Statement

2.1 Physical Setup

The simulation models a stiff block impacting a triangular mesh under gravity. The two bodies in the PolyFEM scene are labelled by convention throughout this thesis:

Body 1: the mesh (deformable). A NeoHookean solid²¹ parameterized by the three design variables (θ, h, E) , namely the apex angle $\theta \in [60^\circ, 110^\circ]$, the wall thickness $h \in \{0.03, 0.035, \dots, 0.07\}$, and the mesh Young’s modulus $E \in [4, 45]$ MPa. Body 1 is the object the surrogate is asked to design and the object on which the gradient surrogate’s objective is evaluated.

Body 2: the impacting block (stiff). A stiff steel block with Young’s modulus 200 GPa, fixed across all simulations. The block is roughly four orders of magnitude stiffer than the mesh, so it stays effectively undeformed and acts as the impactor, but it is still modelled as an elastic body rather than an idealized rigid body. Body 2 is not a design variable, but its peak Von Mises stress is the deployment relevant quantity that the forward surrogate predicts, because it is proportional to the force transmitted to the protected payload in packaging.

Boundary and loading conditions. The mesh (body 1) rests on a fixed rigid ground plane, and the block (body 2) starts just above the mesh and is driven downward under gravity. Frictionless contact is enforced at both the block-mesh and mesh-ground interfaces using a barrier method with barrier distance $\hat{d} = 0.012$. The transient is solved over $t \in [0, 0.5]$ s with a fixed time step $\Delta t = 0.01$ s on a 51-step grid. Material properties are fixed except for the mesh Young’s modulus E :

both bodies use Poisson’s ratio $\nu = 0.45$, the mesh density is $1,100 \text{ kg/m}^3$, and the block density is $7,850 \text{ kg/m}^3$. Only the three design variables (θ, h, E) vary across the dataset; every other simulation setting is held constant so that the learned mapping isolates the effect of the design parameters.

The two surrogate targets are deliberately defined on different bodies, reflecting the two distinct engineering questions. The forward target is the peak Von Mises stress²⁴ on body 2 (the block),

$$\sigma_{\text{peak}}^{\text{block}} = \max_{t \in [0, T]} \sigma_{\text{VM}}^{\text{body 2}}(t),$$

which determines transmitted force, the critical quantity for the protected payload. The gradient targets are the parameter sensitivities of a smooth temporal maximum of the body 1 (mesh) Von Mises stress,

$$L = \text{smooth_max}_t \sigma_{\text{VM}}^{\text{body 1}}(t), \quad \nabla_{(E, h, \theta)} L = (dL/dE, dL/dh, dL/d\theta),$$

obtained from PolyFEM via discrete adjoint differentiation.²² Putting it succinctly: *body 2 stress answers “how hard does the cargo get hit”, and body 1 stress answers “how should the mesh be reshaped to absorb that hit better”.*

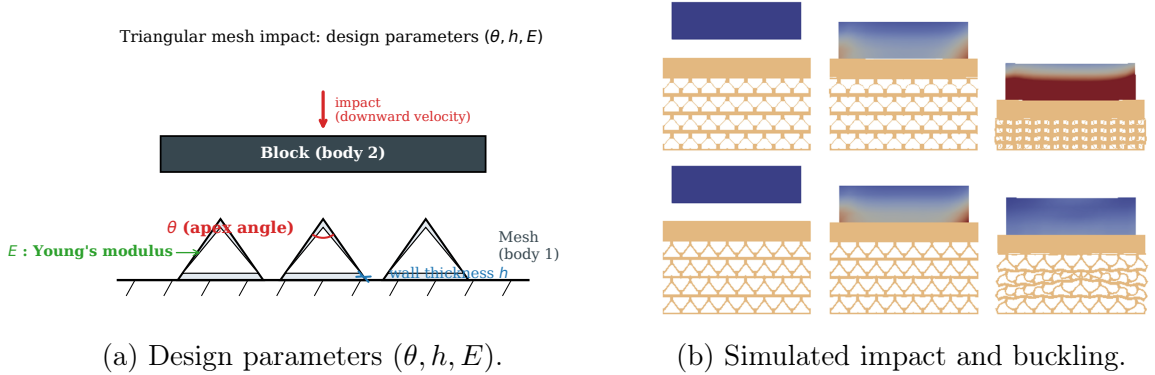


Figure 2.1: Simulation setup. (a) Schematic of the impact: a stiff block (body 2) is driven down onto a deformable triangular mesh (body 1) resting on a fixed ground. The three design parameters (θ, h, E) — apex angle, wall thickness, and Young’s modulus — control mesh geometry and stiffness. (b) Actual PolyFEM renders (columns are successive stages of one impact; color is Von Mises stress). The rightmost frames show the bending dominated collapse of the mesh after buckling, which is the physical origin of the noisy valley studied in Section 2.3.

2.2 Problem Definition

The full impact simulation has a large number of degrees of freedom: the mesh geometry, the material models of both bodies, the loading history, the contact treatment, and the time discretization. Learning a surrogate over all of them at once is neither necessary nor useful, because a packaging designer controls only a few of them in practice. This thesis therefore studies a controlled abstraction of the full problem, and it is worth stating exactly what that abstraction keeps and what it drops.

Three design inputs. The design space is reduced to the three variables a designer actually tunes for this mesh family: the apex angle θ , which sets the cell geometry and controls whether the structure buckles; the wall thickness h , which sets how much material carries the load; and the mesh Young’s modulus E , which sets the base stiffness of that material. These three span the trade-off between a stiff structure that transmits force and a compliant one that absorbs it, which is the design decision that matters for impact protection.

One scalar output. The forward surrogate predicts a single number, the peak Von Mises stress on the block, because in packaging the transmitted peak force is the failure criterion for the protected payload. The full stress field over space and time contains far more information, but the deployment decision only needs its peak, so predicting the scalar directly is both sufficient and easier than predicting the trajectory and extracting its peak afterward (Section 5.3 quantifies this). The gradient surrogate predicts the three sensitivities of the mesh objective L , which are the inputs an optimizer consumes to improve the design.

What is held fixed. Everything outside (θ, h, E) is frozen so that the learned mapping isolates the design levers. The block material and its high stiffness, both Poisson’s ratios, both densities, the frictionless contact model, the single downward impact under gravity, and the NeoHookean elastic material model are all constant. The abstraction therefore excludes plasticity, damage, rate dependence, friction, oblique or repeated loading, and alternative cell topologies. These simplifications bound the scope of the surrogate: it is a fast, accurate map for this mesh family under this loading, not a general impact solver. The limitations chapter (Section 6) revisits which of these would require regenerating data and retraining. With the scope fixed, the two learning problems reduce to a clean forward map $f_{\text{fwd}}: (\theta, h, E) \mapsto \sigma_{\text{peak}}^{\text{block}}$ and a gradient map $f_{\text{grad}}: (\theta, h, E) \mapsto \nabla_{(E,h,\theta)} L$. The remaining sections explain why, despite only three inputs, both maps are hard to learn.

2.3 Forward Challenge: Bifurcation Between Buckling Modes

The block peak stress as a function of θ is not monotonic. Before buckling, the mesh transmits force as a stiff stretching dominated structure and the block experiences high stress. After buckling, the mesh absorbs energy through cell wall bending and the transmitted force drops sharply into a noisy valley before slowly recovering.^{6,8} The Euler buckling load for a strut scales as

$$P_{\text{cr}} \propto \frac{Eh^3}{L_{\text{eff}}(\theta)^2}, \quad L_{\text{eff}}(\theta) = L / \sin \theta,$$

so P_{cr} minimizes near $\theta = 90^\circ$. The mode transition near $\theta \approx 88^\circ$ produces a stress drop of roughly $6\times$ over a $\sim 2^\circ$ window, and the exact location and depth of the valley depend sensitively on h and E . The coefficient of variation of the block peak stress over the dataset is 0.99, an extreme dispersion that any global regressor must learn to navigate.

The vertical spread at any fixed angle in Figure 2.2 is not noise; it is the effect of the other two design variables. Coloring the same scatter by h and by E (Figure 2.3) makes this explicit. Wall thickness h separates the response into ordered bands, thicker walls carrying higher stress, consistent with the Eh^3 scaling of the buckling load, and E shifts the response smoothly within each band. Both variables also move the exact location and depth of the valley, which is why the transition zone is not a single curve but a wide, structured cloud that the surrogate has to resolve from three inputs alone.

2.4 Gradient Challenge: Dynamic Range and Sign Flips

Adjoint gradients of L span six to nine orders of magnitude across the design space, with imbalanced sign distributions that come from the physics (e.g., increasing wall thickness almost always decreases the loss) (Table 2.1). A single MSE loss on raw values is dominated by the largest magnitudes, so small gradients become invisible during training. The same bifurcation that complicates the forward task reappears in derivative form. Gradients are continuous in smooth regions but jump in sign and

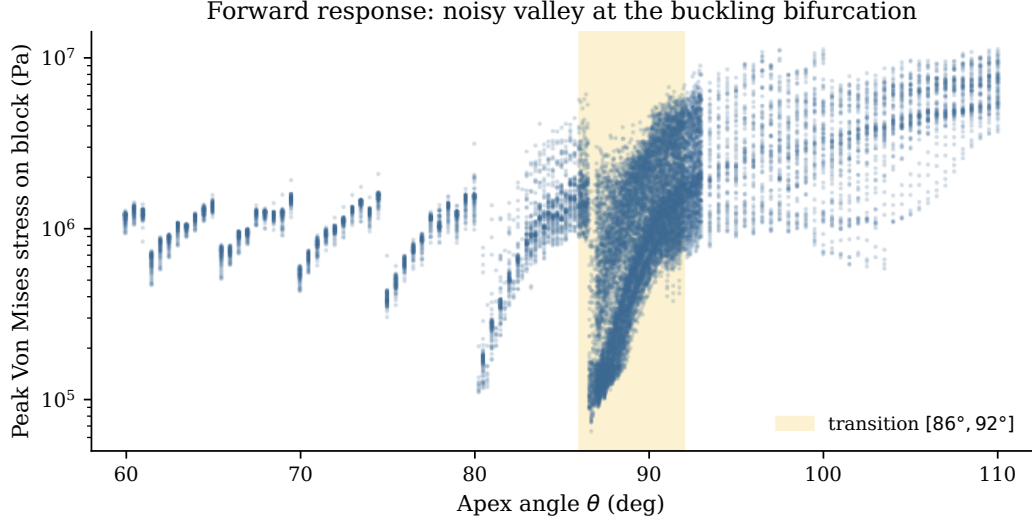


Figure 2.2: Each dot is one simulation: the impacting block’s peak Von Mises stress (vertical axis) plotted against the mesh apex angle θ (horizontal axis), for all 18,887 valid samples. Different vertical positions at the same θ come from different combinations of wall thickness h and Young’s modulus E . For $\theta < 86^\circ$ the response is high and stable. Near $\theta \approx 88^\circ$ (the highlighted band) the mesh switches from a stretching dominated mode to a bending dominated mode and the stress collapses sharply, with large scatter that depends on h and E . The response recovers only partially for larger θ . This noisy valley is the main reason a single global regressor struggles in this region.

magnitude at the mode transition. The FEM solver’s Newton iteration also takes different convergence paths on either side of the bifurcation, which adds noise that no amount of simulation refinement can remove from the supervision signal for $\theta > 91^\circ$.

Table 2.1: Adjoint gradient ranges and sign skew across the design space.

Gradient	Range	Sign distribution
dL/dE	$[-2.5 \times 10^{14}, 2.2 \times 10^{13}]$	balanced
dL/dh	$[-1.3 \times 10^{18}, 1.0 \times 10^{17}]$	86% negative
$dL/d\theta$	$[-5.2 \times 10^{14}, 9.7 \times 10^{14}]$	84% positive

The scatter at a fixed angle is explained by h and E

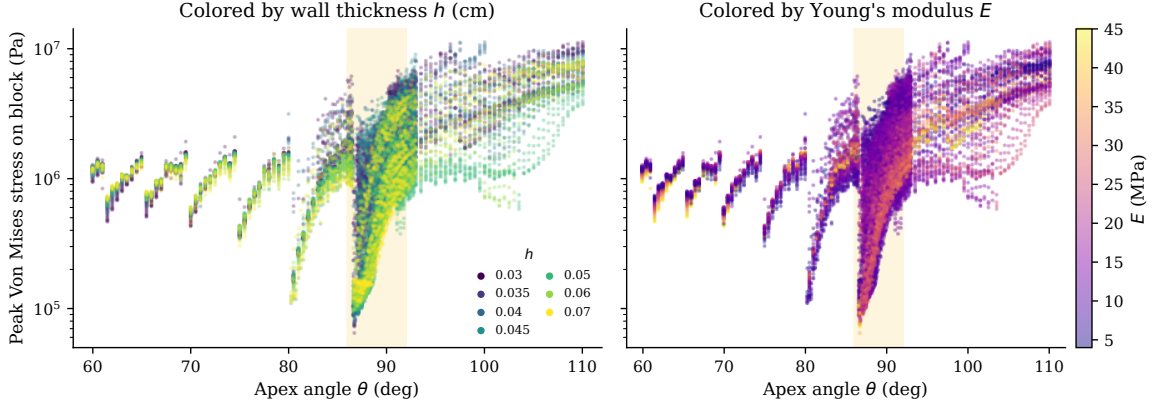


Figure 2.3: The same peak-stress-versus- θ scatter as Figure 2.2, color-coded by the other two design variables. Left: colored by the seven discrete wall thicknesses h ; thicker walls form higher, ordered stress bands. Right: colored by Young's modulus E on a continuous scale. The vertical scatter at any fixed θ , including inside the transition band, is largely explained by h and E rather than being random. A two dimensional scatter with a color dimension is used in place of a three dimensional scatter, which is difficult to read on paper.

Gradient targets span many decades and switch sign

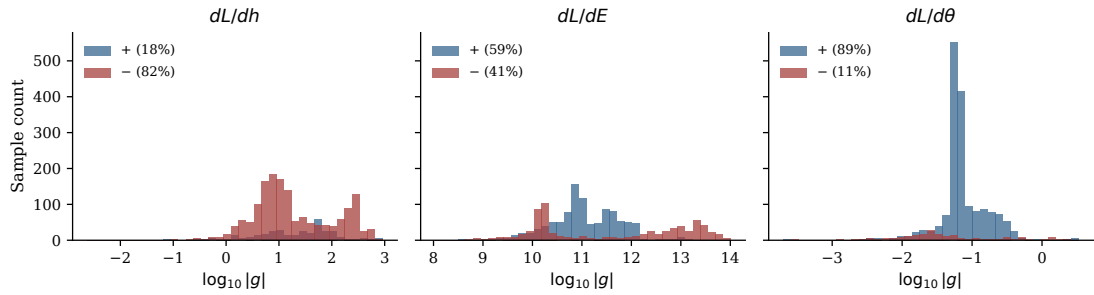


Figure 2.4: Log-magnitude histograms of the three adjoint gradients, colored by sign. Each gradient spans 6–9 orders of magnitude; sign distributions are imbalanced (dL/dh : 86% negative; $dL/d\theta$: 84% positive). A single MSE loss cannot treat all samples equally.

Chapter 3

Related Work

This chapter positions the thesis against four bodies of work: machine learning surrogates for the finite element method, the comparison between tree ensembles and deep learning on tabular data, neural operators, and differentiable simulation with the adjoint method. A common thread is that most existing surrogate work assumes a smooth mapping from design parameters to response, whereas the problem studied here is shaped by a structural bifurcation. That gap is what this thesis addresses.

ML surrogates for FEM. The use of machine learning models as surrogates for the finite element method has a growing literature. Kudela and Matousek¹⁵ survey over one hundred papers on surrogate models for FEM computations, noting that the majority focus on static or quasistatic linear elastic problems with smooth mappings from parameters to response, and that the accuracy metrics relevant for deployment remain underexplored. Badarinath and Chierichetti² demonstrated supervised learning as an FEM surrogate for one-dimensional dynamic systems, achieving error below one percent on stress distributions and below ten percent on peak stresses; their test cases remain smooth and well conditioned. Gongora et al.⁹ applied ML surrogates to lattice design with $R^2 > 0.95$ and roughly 85% of predictions within 5% relative error, reducing the time per evaluation from 37 minutes to under one second. Their lattice response is smooth and does not exhibit the mode switching that makes our problem hard in the present work. A common limitation in the existing surrogate literature is the assumption that target functions are smooth over the design space. When this assumption breaks down near structural bifurcations, contact events, or mode transitions, reported R^2 values can mask large localized errors. This thesis tackles that less explored regime.

Trees versus deep learning on tabular data. Grinsztajn et al.¹⁰ estab-

lished, across forty-five datasets and twenty-thousand GPU hours of tuning, that tree ensembles consistently outperform deep learning on medium scale tabular data. Their analysis attributes this to three properties: trees ignore uninformative features through information gain splitting, they capture axis aligned decision boundaries with a single split, and they handle irregular, nonsmooth functions natively. The forward problem in this thesis exhibits all three: discrete h levels, an axis aligned θ threshold near the bifurcation, and sharp nonmonotonic behavior. Section 5.3 is consistent with these predictions. Tuned XGBoost reaches 74.1% Acc@10% while a 114K-parameter deep MLP and a tabular Transformer reach only 45.3% and 50.7%. The finding here is not that trees are universally superior, but that for the specific structure of the mesh problem (three inputs, a scalar output, and a transition aligned with an axis), tree ensembles are well suited.

Neural operators. Lu et al.¹⁷ introduced DeepONet for learning nonlinear operators between function spaces, proving a universal approximation theorem for operators. Kovachki et al.¹⁴ formalized the broader neural operator class, and Li et al.¹⁶ proposed the Fourier Neural Operator (FNO) as a resolution invariant spectral method for PDE surrogate modeling. A comprehensive comparison by Lu et al.¹⁸ across sixteen benchmark problems found that FNO achieves lower in distribution error on regular grids, while DeepONet is substantially more robust to input noise and distribution shift. FNO error increases by a factor of 10,000 at 0.1% noise in one hypersonic boundary layer benchmark. Azizzadenesheli et al.¹ provide a review of neural operators for scientific simulation and design, noting that operator learning methods excel when the target is a smooth function or field. He et al.¹¹ extend DeepONet to sequential prediction of full field solutions under time dependent loads, achieving $R^2 > 0.99$ on solid mechanics dependent on path with smooth loading histories. Neural operators are evaluated here as Method 1 (predict the full transient stress trajectory, then extract the peak). The key finding is that the operator’s strong trajectory fit ($R^2 = 0.92$) does not translate to accuracy on the extracted peak (39.5% Acc@10%) because small errors at each timestep compound near the minimum of the noisy valley where the peak occurs.

Differentiable simulation and adjoint methods. The adjoint method computes gradients of simulation objectives with respect to design parameters at a cost independent of the parameter dimension, making it the standard approach for optimization constrained by PDEs.²² In the discrete adjoint formulation used here, PolyFEM differentiates the fully discretized system (including contact resolution and

Newton iteration) to obtain exact gradients of the converged solution. The resulting gradients are exact up to floating point arithmetic but can be noisy when the forward solver’s convergence path is sensitive to small parameter perturbations. This is precisely the situation that arises in the region after the mesh has buckled in the mesh problem. Recent work on differentiable simulation frameworks (e.g., DiffTaichi, JAX-FEM, and PolyFEM’s own adjoint interface) has made adjoint gradients accessible as drop in components in optimization pipelines. The cost remains that of a full forward and backward simulation pass (~ 30 minutes per sample in the present setting), motivating the gradient surrogate developed in this thesis.

Sobolev training. Czarnecki et al.⁵ introduced Sobolev training, jointly supervising a neural network on function values and their derivatives. The gradient supervision acts as a regularizer that encourages the model’s learned function to match both the target and its local slopes. In the present work, Sobolev training is implemented as a baseline for the gradient surrogate and is found to improve sign agreement modestly (from 87% to 89%). Its limitation is fundamental. When gradients span six to nine orders of magnitude, a single MSE loss on raw gradient values is dominated by the largest targets, and the model cannot simultaneously learn the sign structure of samples with small magnitude. The proposed decomposition into sign and the logarithm of magnitude addresses three conflicts that Sobolev training does not: *scale conflict* (log-space regression treats all magnitudes equally), *task conflict* (classification for sign and regression for magnitude admit different optimal losses), and *capacity conflict* (each subtask receives dedicated head capacity). This decomposition is not limited to neural networks. It is applied here to both XGBoost and a multihead NN, demonstrating that it holds across model families.

Chapter 4

Methodology

This chapter develops the methodology for both surrogates. Section 4.1 covers the forward surrogate (the Method 1 versus Method 2 framing, model selection, feature engineering, target transform, and hyperparameters). Section 4.2 covers the gradient surrogate (sign and log-magnitude decomposition, multihead architecture, loss function, and training). Evaluation metrics used in the empirical chapter are defined in Section 5.1.

4.1 Forward Surrogate

Two prediction approaches. *Method 1 (Time-series $\rightarrow$ Peak)* predicts the full 51-step Von Mises stress trajectory $\hat{\mathbf{y}}(t) \in \mathbb{R}^{51}$ and extracts $\hat{y}_{\text{peak}} = \max_t \hat{\mathbf{y}}(t)$. Models considered: MLP and DeepONet,^{14,17} evaluated by 5-fold cross validation. *Method 2 (Direct Peak)* predicts $\hat{y}_{\text{peak}}$ directly from (θ, h, E) . Models considered: Random Forest,³ XGBoost,⁴ GBR,⁷ SVR, MLP, evaluated on a stratified 80/10/10 train/validation/test split. The Method 1 versus Method 2 contrast is central to the experimental design. Predicting time series yields rich diagnostic information but introduces $51\times$ more output dimensions, temporal correlations, and error compounding at peak extraction. The noisy valley minimum amplifies any error at a single timestep.

Model selection for Method 2. The input is 3D tabular with continuous (θ, E) and discrete $(h: 7 \text{ values})$ variables, and the target spans 2.2 orders of magnitude. Tree ensembles are favored a priori.¹⁰ Their axis aligned splits capture the discrete h structure and the $\theta \approx 88^\circ$ threshold natively. Models excluded: k -NN is unlikely to

extrapolate well within sparse transition regions; Gaussian Processes scale as $O(n^3)$ exact or $O(nm^2)$ sparse, impractical at $n = 18,887$; CNNs and Transformers are designed for spatial or sequential locality that is largely absent in 3D tabular data.

From exploration to features. The feature set was not designed in advance; it grew out of exploratory analysis of the forward dataset, and each group of features answers a specific thing that analysis revealed. Plotting peak stress against the apex angle (Figure 2.2) showed that the response is flat and high for small θ , then collapses into a noisy valley near $\theta \approx 88^\circ$ before partially recovering. This is the buckling bifurcation, and it is the single hardest feature of the data, so the model is given the distance to it directly, as $|\theta - 88|$ and its powers, rather than being left to discover a sharp axis aligned threshold from the raw angle alone. Inspecting the sampling of the three inputs (Figure 5.1) showed that h is not continuous but takes seven discrete manufacturing thicknesses, which motivated a one-hot encoding so the model treats each thickness as its own level instead of interpolating between them. Looking at the target range showed peak stress spanning more than two orders of magnitude, which motivated the $\log(1 + y)$ transform discussed below. The remaining features come from dimensional analysis of cellular solid mechanics rather than from the data directly: the bending stiffness proxies h^2E and h^3E follow the plate-bending scaling, and the pairwise interactions let a tree combine two inputs in a single split. Feature importance was then checked with SHAP¹⁹ to confirm each group carried signal, and the ablation in Section 5.3 confirms the features, taken together, are what make the surrogate usable.

Feature engineering. Feature design is guided by dimensional analysis of cellular solid mechanics^{6,8} and validated via SHAP importance.¹⁹ The full 22-dimensional

feature vector is grouped by physical role:

raw inputs:	θ, h, E
log compression:	$\log h, \log E$
geometry encoding:	$\sin(\theta\pi/180)$
transition proximity:	$ \theta - 88 , (\theta - 88)^2, (\theta - 88)^3$
bending stiffness proxies:	h^2E, h^3E
compliance:	h/E
pairwise interactions:	$\theta h, \theta E, hE$
discrete-thickness one-hot:	$\mathbf{1}[h = h_k], \quad k = 1, \dots, 7.$

The bending stiffness terms reflect the plate-bending scaling $D \propto Eh^3/12(1 - \nu^2)$. The transition proximity terms encode distance to the buckling bifurcation at $\theta \approx 88^\circ$ in both symmetric and asymmetric forms. The discrete one-hot prevents spurious interpolation between adjacent thickness levels.

Target transform. Peak stress spans 6.5×10^4 to 1.1×10^7 Pa. Applying $\tilde{y} = \log(1 + y)$ aligns training with Acc@K%: for $y \gg 1$,

$$\log y - \log \hat{y} = \log \frac{y}{\hat{y}} \approx \frac{y - \hat{y}}{y},$$

so MSE in log space approximately minimizes squared relative error.²⁵ A sample with 10% relative error contributes $(\log 1.1)^2 \approx 0.009$ regardless of whether $y = 100$ K or $y = 10$ MPa, which is the scale invariance Acc@K% requires.

Hyperparameters. XGBoost: 5,000 trees, max depth 8, learning rate 0.01, subsample 0.8, column subsample 0.7. Random Forest: 3,000 trees, fully grown, max_features = 0.5. Hyperparameters were selected on the validation fold. The test fold was evaluated only once.

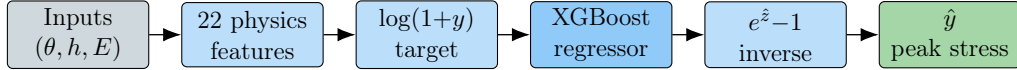


Figure 4.1: End to end forward surrogate pipeline: raw inputs (θ, h, E) are expanded to 22 features informed by physics; the target is $\log(1 + y)$ -transformed; XGBoost regresses in log space; the inverse transform yields the stress prediction.

4.2 Gradient Surrogate

Sign and log-magnitude decomposition. The primary methodological contribution is a decomposition of each gradient prediction into two subtasks (sign and shifted log-magnitude) coupled with a value prediction auxiliary head:

$$\hat{g} = \hat{s} \cdot (10^{\hat{m}} - 1), \quad \hat{s} \in \{-1, +1\}, \quad \hat{m} \approx \log_{10}(|g| + 1).$$

The +1 shift in the magnitude target is a numerical stabilizer that keeps small gradients (where $|g| \rightarrow 0$) from producing $\log_{10} |g| \rightarrow -\infty$ during training. It is inverted at decoding time. The sign $\hat{s}$ is produced by a binary classifier with binary cross entropy loss. The log-magnitude $\hat{m}$ is produced by a regressor with MSE loss in $\log_{10}$ space. The decomposition resolves three conflicts that a single MSE loss on raw gradients cannot:

Scale conflict. Log space magnitude regression treats a 10^{15} and a 10^{10} gradient equally, while raw MSE is dominated by the largest values.

Task conflict. Classification (sign) and regression (magnitude) admit different optimal losses, and collapsing them into one MSE forces a compromise that serves neither.

Capacity conflict. Each subtask receives dedicated head capacity, while a single output competes for the same parameters.

The decomposition holds across model families. Section 5.4 evaluates it with both XGBoost⁴ (a separate sign classifier and magnitude regressor) and a multihead neural network.

Multihead architecture. The 3-head variant used for dL/dE shares a 6-stage Linear+SiLU backbone of hidden width 512 across three Linear-SiLU-Linear heads:

$$f_\phi: (\theta, h, E) \xrightarrow{\text{z-score}} \mathbb{R}^{512} \longrightarrow (\hat{v}, \hat{m}, \hat{s}),$$

$$\hat{v} \approx \log(1+L), \quad \hat{m} \approx \log_{10}(|g|+1), \quad \hat{s} = \sigma(\cdot).$$

For $[dL/dh, dL/d\theta]$ the architecture extends to 5 heads: a shared value head plus $(\text{mag}_h, \text{sign}_h, \text{mag}_\theta, \text{sign}_\theta)$, all of identical Linear-SiLU-Linear shape. Each head’s hidden width matches the backbone width (512). Final predictions are taken from

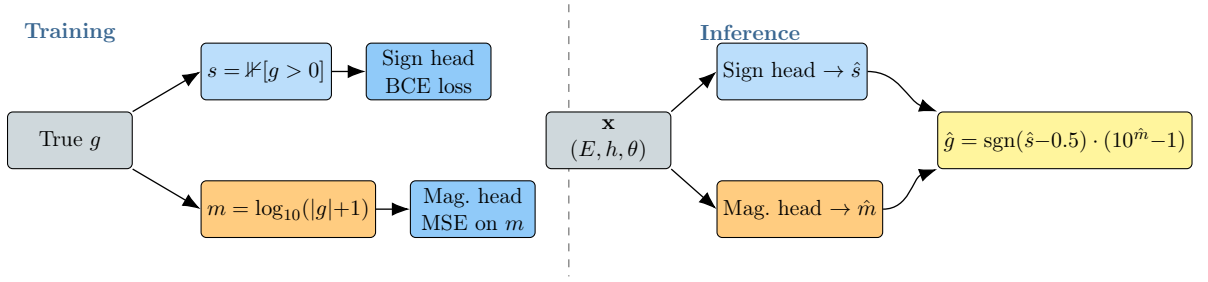


Figure 4.2: Decomposition into sign and the logarithm of magnitude. Training: the true gradient g is split into a binary sign target and a log-magnitude target, supervised by separate losses. Inference: the two heads’ outputs are recombined into the final gradient prediction.

a single model trained per outer cross validation fold (no seed ensembling within a fold). The full architecture is shown in Figure 4.3.

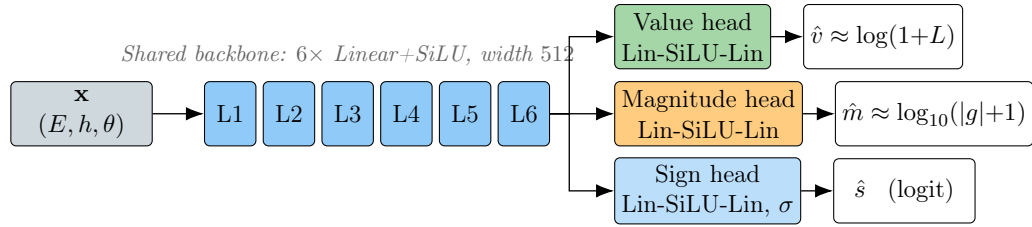


Figure 4.3: Multihead architecture (3-head version, used for dL/dE). The shared backbone of six Linear+SiLU stages feeds three Linear-SiLU-Linear heads: a value head, a sign head (sigmoid output), and a magnitude head ($\log_{10}(|g|+1)$). For $[dL/dh, dL/d\theta]$ the architecture extends to five heads (one {mag, sign} pair per gradient component, plus the shared value head).

Loss function. For dL/dE :

$$\mathcal{L} = \text{MSE}(\hat{v}, v) + \lambda_{\text{ramp}}[3.0 \mathcal{L}_{\text{mag}} + 1.5 \text{BCE}(\hat{s}, s)].$$

For $[dL/dh, dL/d\theta]$:

$$\mathcal{L} = \text{MSE}(\hat{v}, v) + \lambda_{\text{ramp}}[2.0 \mathcal{L}_{\text{mag}}^h + 1.0 \text{BCE}^h + 3.0 \mathcal{L}_{\text{mag}}^\theta + 1.5 \text{BCE}^\theta],$$

with $\lambda_{\text{ramp}} = \min(1, \max(0, (\text{epoch} - 30)/40))$ to stabilize the backbone before the gradient heads activate. Magnitude loss is MSE for all three gradients in the canonical proper cross validation protocol.

Training. AdamW with learning rate 5×10^{-4} and weight decay 10^{-5} ; cosine annealing to $T_{\text{max}} = 400$ epochs; batch size 512; gradient clipping at $\|\cdot\|_2 \leq 1$; early

stopping on Acc@20% measured on the inner validation set, checked every 20 epochs with patience 40. Each gradient component is evaluated under rigorous 5-fold cross validation, with no seed ensembling at the headline level. The dataset is partitioned into 5 outer folds (seed 42). Within each fold a single model is trained (seed $42 + i$, $i = 0, \dots, 4$). The training fold is split 90/10 into an inner training set and an inner validation set used solely for early stopping, so the test fold is never seen during model selection. This avoids the roughly 5 pp Acc@20% inflated accuracy that occurs when early stopping monitors the test set directly.

Chapter 5

Experiments

5.1 Evaluation Metrics

Headline metrics. Throughout the experiments the forward surrogate is judged by **Acc@10%** (fraction of test samples whose prediction lies within 10% of the true value), and the gradient surrogate by **Acc@20% combined with Sign%** (fraction whose sign matches the true gradient). R^2 , MAPE, and $R_{\log}^2$ are reported alongside, for comparability with existing FEM surrogate work and as diagnostics, but are not the deployment indicator. The choice of these as headline metrics, and why R^2 on its own is misleading for this problem, is explained below.

Metric definitions. For forward regression, given n samples with true values $\{y_i\}$ and predictions $\{\hat{y}_i\}$:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad \text{MAPE} = \frac{100\%}{n} \sum_i \frac{|y_i - \hat{y}_i|}{|y_i|},$$

$$\text{Acc@}K\% = \frac{100\%}{n} \sum_i \mathbf{1}\left(\frac{|y_i - \hat{y}_i|}{|y_i|} \leq \frac{K}{100}\right).$$

For gradient regression, sign agreement and $R_{\log}^2$ supplement the above:

$$\text{Sign}\% = \frac{100\%}{n} \sum_i \mathbf{1}(\text{sign}(\hat{g}_i) = \text{sign}(g_i)), \quad R_{\log}^2 = 1 - \frac{\sum_i (\log_{10} |\hat{g}_i| - \log_{10} |g_i|)^2}{\sum_i (\log_{10} |g_i| - \overline{\log_{10} |g|})^2}.$$

Why Acc@K% as the primary metric. A regression model outputs continuous predictions, but the engineering question this thesis addresses is binary: *can this prediction safely replace FEM?* Acc@K% answers that question directly by counting

how many predictions land within a tolerance band. The forward threshold $K = 10$ matches the engineering tolerance for structural stress, which sits well within the 1.5– $3\times$ safety factors used in practice.^{2,9} The gradient threshold $K = 20$ is chosen so that a 20% magnitude error in the gradient still satisfies Armijo descent under line search, so optimisation still converges, only with a slightly suboptimal step. $\text{Acc}@K\%$ is also robust to outliers (a single catastrophic miss reduces $\text{Acc}@10\%$ by only $1/n$) and is scale invariant, treating a 10^{15} gradient and a 10^{10} gradient equally.

Why Sign% for gradients. A gradient with the wrong sign moves the optimizer away from the optimum regardless of how accurate its magnitude is. Sign agreement is therefore the binary diagnostic for the gradient surrogate, paired with $\text{Acc}@20\%$ to also quantify the magnitude error.

Why also report R^2 . R^2 is the standard goodness-of-fit number reported in the surrogate modeling literature¹⁵ and is required for any comparison to existing FEM surrogate results. Reporting it alone, however, is misleading for this problem. Kudela and Matousek¹⁵ note that high R^2 does not imply local fidelity, and Jones et al.¹² make the same point in the context of Bayesian optimization. Section 5.3 makes it concrete: the XGBoost test set reaches $R^2 = 0.97$ while 26% of predictions are still outside the 10% tolerance, a gap that the aggregate R^2 hides. MAPE and $R^2_{\log}$ are reported in the same diagnostic spirit. The $\text{Acc}@K\%$ plus Sign% combination is therefore the deployment indicator throughout, with the aggregate metrics retained for context and explicitly contrasted whenever they diverge.

5.2 Datasets

Simulation pipeline. All FEM data are generated using PolyFEM²² with its Python interface. Each simulation models the impact of a stiff steel block (body 2; $E = 200$ GPa fixed, $\nu = 0.45$, $\rho = 7,850$ kg/m³) onto a deformable triangular mesh (body 1; E variable, $\nu = 0.45$, $\rho = 1,100$ kg/m³, NeoHookean²¹). Contact is frictionless with a barrier distance $\hat{d} = 0.012$. The transient is solved over $t \in [0, 0.5]$ s with $\Delta t = 0.01$ s (51 time steps). For the gradient dataset, PolyFEM’s adjoint solver computes exact discrete gradients of the smooth-max Von Mises objective on body 1 with respect to (E, h, θ) in a single backward pass after the forward solve. A single forward solve takes on the order of minutes of CPU time, and a combined forward and adjoint pass takes meaningfully longer. The full data generation campaign runs on Compute Canada CPU resources as an offline batch job.

Parameter space and sampling. The nonuniform sampling grid reflects physical structure rather than uniform coverage. Table 5.1 summarises the per parameter strategy. The apex angle θ is densely sampled in the buckling transition window and coarser outside. Wall thickness h is a small set of discrete values, with the gradient dataset using a strict subset of the forward dataset’s grid. Young’s modulus E has an h dependent value count, because thinner walls tolerate higher E without numerical instability while thicker walls buckle earlier. The dense sampling in the transition region is essential. Section 5.3 shows that with the standard grid the surrogate’s leave- θ -group-out Acc@10% is 50.4% versus 73.4% on standard cross validation, indicating heavy reliance on interpolation between neighboring θ values.

Table 5.1: Per parameter sampling strategy for the FEM data generation.

Parameter	Region / Dataset	Range and step	# values
θ (deg)	Smooth [60, 86]	$\Delta\theta = 0.5$	53
	Transition [86, 92]	$\Delta\theta = 0.05\text{--}0.1$	101
	Post transition [93, 110]	$\Delta\theta = 0.5$	35
h (wall thickness)	Forward dataset	$\{0.03, 0.035, 0.04, 0.045, 0.05, 0.06, 0.07\}$	7
	Gradient dataset	$\{0.03, 0.04, 0.05, 0.06, 0.07\}$	5
E (MPa)	Range	[4, 45]	up to 33 per h

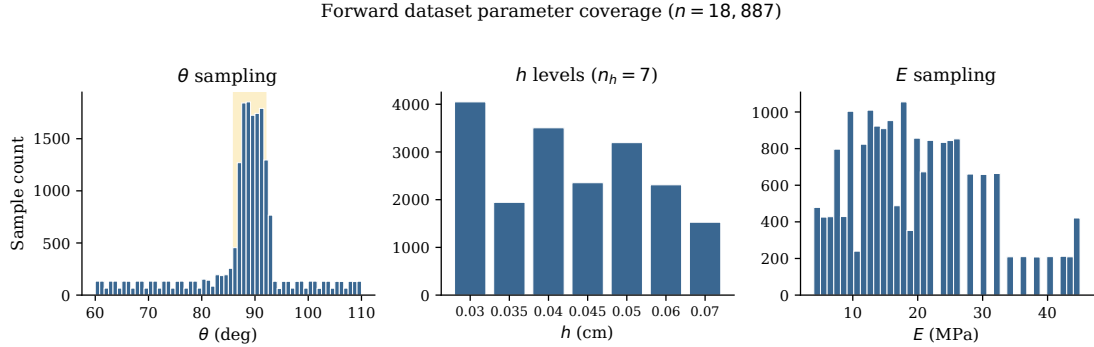


Figure 5.1: Parameter sampling for the forward dataset ($n = 18,887$). Left: θ is densely sampled in the $[86^\circ, 92^\circ]$ transition zone. Centre: h takes 7 discrete levels. Right: E coverage in $[4, 45]$ MPa.

Datasets. Three datasets are used, summarised in Table 5.2. The forward dataset (Experiment 1) records the peak Von Mises stress on body 2 at each PolyFEM transient solve. The two gradient datasets (Experiment 2) record the discrete adjoint sensitivities of the smooth-max Von Mises objective on body 1 with respect to material (dL/dE) and geometric (dL/dh , $dL/d\theta$) parameters. The geometric gradients

additionally require the chain rule $dL/dX \cdot dX/d[h, \theta]$ through the mesh parameterization.

Table 5.2: Datasets, target ranges, and sign skew. Sample counts are after filtering numerical failures.

Dataset	Target	n	Range	Sign
Forward (Exp. 1)	$\sigma_{\text{peak}}^{\text{block}}$	18,887	$[6.47 \times 10^4, 1.10 \times 10^7]$ Pa (CV 0.99)	—
Material (Exp. 2a)	dL/dE	8,606	$[-2.5 \times 10^{14}, 2.2 \times 10^{13}]$	$\approx 50/50$
Geometric (Exp. 2b)	dL/dh	9,653	$[-1.3 \times 10^{18}, 1.0 \times 10^{17}]$	86% negative
	$dL/d\theta$		$[-5.2 \times 10^{14}, 9.7 \times 10^{14}]$	84% positive

Splits and cross validation. The forward surrogate uses an 80/10/10 stratified split producing 15,109 training, 1,889 validation, and 1,889 test samples. Stratification is by five θ regions ($< 80^\circ$, $[80, 87)$, $[87, 93)$, $[93, 100)$, $\geq 100^\circ$) to ensure proportional representation of the bifurcation zone. The test set is kept aside during all hyperparameter selection and evaluated exactly once. Robustness is verified via 5-fold stratified cross validation.¹³ The gradient surrogate uses a 60/20/20 stratified split for the XGBoost baselines (train / validation for early stopping / final test) with stratification by $h \times \text{sign}(g)$. The multihead neural network uses rigorous 5-fold cross validation with the same stratification scheme. Within each training fold, an inner 90/10 random split provides the validation set used for early stopping, so the test fold of each outer split is never seen during model selection. This protocol avoids the ~ 5 pp Acc@20% inflated accuracy seen when early stopping uses test data.

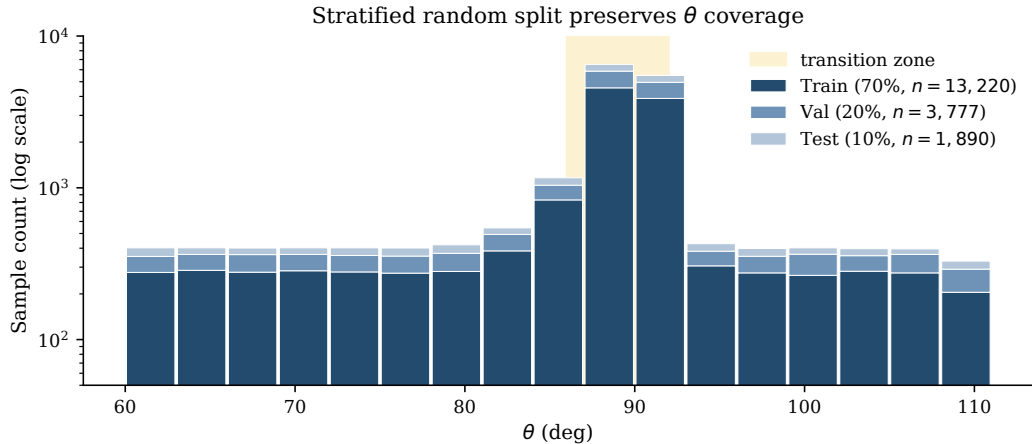


Figure 5.2: Stratified random split (70/20/10) preserves θ coverage across all three partitions. The highlighted band marks the densely sampled transition zone.

Compute and reproducibility. FEM simulations were executed on Compute

Canada CPU resources. Surrogate training was performed on Apple M-series silicon. XGBoost training takes approximately 12.3s for the full forward pipeline. The multihead NN ensemble (five models, 800 epochs) takes approximately 45 minutes. Inference for a single sample is 0.03 ms (XGBoost) or under 0.1 ms (multihead NN). All randomness is controlled by fixed seeds (seed 42 for the forward surrogate and the gradient XGBoost baseline; seeds 0–4 for the multihead NN ensemble).

5.3 Forward Surrogate Results

Baseline versus optimized models. Table 5.3 shows Method 2 model performance on the validation set with the three raw inputs (θ, h, E). Random Forest leads on Acc@10% with raw features; all three tree models (RF, XGBoost, GBR) substantially outperform SVR and MLP, consistent with the prediction from Grinsztajn et al.¹⁰ that tree ensembles tend to outperform other model classes on medium scale tabular regression with axis aligned structure.

Table 5.3: Method 2 baseline models (3 raw features), validation set.

Model	R^2	MAPE	Acc@10%	Acc@20%	Time
XGBoost	0.971	14.6%	58.2%	79.2%	0.9 s
RF	0.970	11.4%	69.3%	84.9%	0.6 s
GBR	0.966	16.8%	51.6%	75.0%	2.7 s
SVR	0.841	43.8%	27.2%	49.0%	3.2 s
MLP	0.751	81.8%	14.6%	31.2%	7.9 s

Adding the 22 features informed by physics and the log1p target transform changes the ranking: XGBoost overtakes RF at 74.1% Acc@10% (Table 5.4). The asymmetric gain pattern, in which boosting methods improve dramatically (XGBoost +15.9 pp, GBR +21.3 pp) while bagging improves only modestly (RF +4.0 pp), is consistent with the theoretical distinction^{3,4} that deep RF trees can discover boundaries through recursive splitting while shallow boosting trees rely on explicit residual features.

Table 5.4: Method 2 optimized models (22 features + log1p), validation set.

Model	R^2	MAPE	Acc@10%	Acc@20%	Time
XGBoost	0.973	10.2%	74.1%	86.8%	12.3 s
RF	0.969	10.5%	73.3%	87.5%	11.0 s
GBR	0.971	10.7%	72.9%	86.4%	71.6 s

Final test performance. The final XGBoost model (22 features + log1p) achieves the test set results shown in Table 5.5. The single split test value of 74.1% is reproduced within 1σ ($73.4\% \pm 0.5\%$) by 5-fold stratified cross validation, and matches the validation result exactly, suggesting that the test set was kept clean throughout model selection.

Table 5.5: Final XGBoost test set results.

Metric	Test ($n = 1,889$)
R^2	0.9684
MAPE	10.2%
Acc@10%	74.1%
Acc@20%	86.9%

Where the surrogate fails. Aggregate 74.1% Acc@10% hides a sharply bimodal quality distribution. Decomposing by θ region (Table 5.6) shows nearly perfect prediction in the smooth region before the bifurcation (98.9% for $\theta < 80^\circ$) and a sharp drop to 59.6% in the transition zone $[87^\circ, 89^\circ)$, where 458 test samples sit on the noisy valley. The transition zone R^2 of 0.85 indicates that the regression problem itself becomes substantially harder at the bifurcation, not that the model is overfit elsewhere.

Table 5.6: Regional decomposition of test set performance by apex angle.

Region	n	Acc@10%	Acc@20%	R^2
$\theta < 80^\circ$	268	98.9%	99.6%	0.992
$[80^\circ, 87^\circ)$	186	83.3%	94.6%	0.967
$[87^\circ, 89^\circ)$	458	59.6%	74.7%	0.850
$[89^\circ, 93^\circ)$	742	69.3%	85.3%	0.922
$[93^\circ, 100^\circ)$	96	77.1%	93.8%	0.958
$\theta \geq 100^\circ$	139	85.6%	95.7%	0.974

Why R^2 stays high while Acc@10% drops. It is worth being precise about how a test set R^2 of 0.97 coexists with 26% of predictions failing the 10% tolerance, because the two numbers measure different things. R^2 is one minus the ratio of residual variance to total variance, so it is dominated by the samples that contribute most to the total variance, namely the high-stress samples before the bifurcation, where peak stress reaches 10^7 Pa. Getting those large values approximately right, together with the overall trend from the high plateau down into the valley, already explains most of the variance and pushes R^2 close to one. Acc@10% asks a different

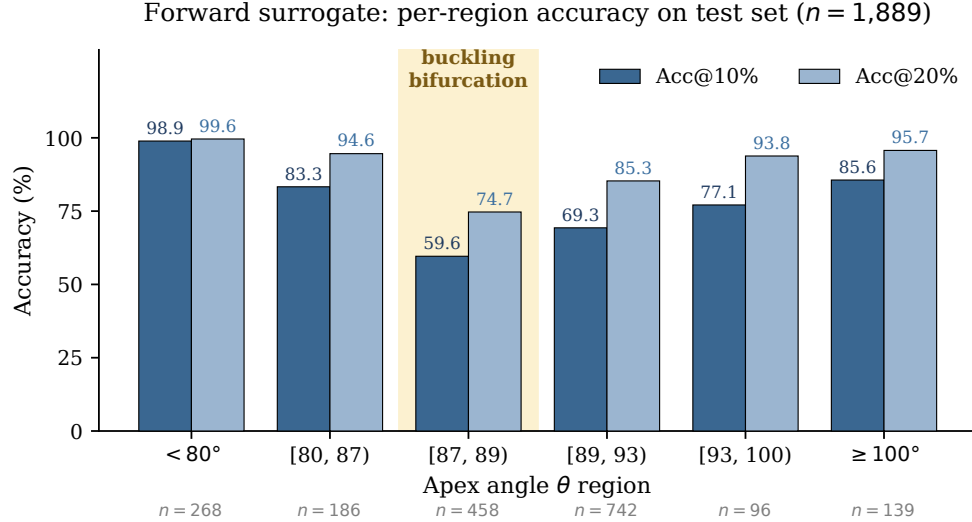


Figure 5.3: Forward surrogate Acc@10% and Acc@20% by apex angle region on the test set ($n = 1,889$). The shaded band marks the transition zone of the bifurcation between buckling modes, where accuracy drops sharply.

and stricter question: it is a per-sample *relative* criterion, counting a prediction as correct only if it lands within 10% of that specific sample’s true value. In the transition zone the true stress has collapsed into the noisy valley and is small, so a modest absolute error becomes a large relative error and the sample fails Acc@10% while barely moving the variance ratio that R^2 tracks. This is exactly the pattern in Table 5.6: the transition zone still has $R^2 = 0.85$, which looks acceptable, but only 59.6% Acc@10%. The same mechanism explains why the weaker baselines in Table 5.3 (SVR at $R^2 = 0.84$, MLP at $R^2 = 0.75$) collapse to 27% and 15% Acc@10%: they capture the broad high-versus-low trend well enough for a respectable R^2 , but are rarely within 10% of any individual value. Aggregate variance explained and per-sample usability are therefore not interchangeable, and for a deployment decision it is the latter that matters.

Leave- θ -group-out generalization. To test whether the surrogate genuinely learns the stress- θ relationship or merely interpolates between dense training points, θ values are partitioned into five disjoint groups; each fold withholds 20% of θ values entirely. The XGBoost model drops from 73.4% standard CV Acc@10% to 50.4% leave- θ -out Acc@10%, a -23.0 pp gap. RF behaves identically (-20.3 pp). This result supports the dense transition region sampling strategy described above. The $\Delta\theta = 0.05^\circ$ grid in $[86^\circ, 92^\circ]$ is required for acceptable accuracy, not oversampling. The surrogate is reliable only for θ values within the training grid.

Method 1 versus Method 2. DeepONet^{14,17} achieves excellent fit over the time series ($R^2 = 0.92$) but only 39.5% Acc@10% on the extracted peak (Table 5.7), which is 34.6 pp below the direct scalar predictor. The gap is likely a problem of task formulation, not a limit of the neural operator. Predicting time series adds $51\times$ more output dimensions, temporal correlations, and error compounding at peak extraction near the noisy valley minimum. Predicting an intermediate quantity when the target is directly available appears to compound error unnecessarily.

Table 5.7: Method 1 (time series to peak) results, 5-fold CV.

Model	TS R^2	Peak R^2	Peak MAPE	Peak Acc@10%	Peak Acc@20%
DeepONet	0.916	0.941	22.3%	39.5%	66.0%
MLP	0.698	0.820	124.3%	22.9%	41.2%

Forward ablation. Removing the 22 physics features (keeping log1p) drops accuracy by 31.2 pp, far more than removing log1p alone (-3.7 pp) (Table 5.8). Notably, log1p *without* features (42.9%) is worse than MSE *without* features (58.2%). Log compression amplifies confusion among already similar transition samples in the low stress range. Only after features separate them does the alignment of log1p with relative error yield gains. Features and log1p are synergistic, not additive.

Table 5.8: Forward ablation, validation set.

Configuration	Acc@10%	Δ vs. Full
Full pipeline (22 feat + log1p + XGBoost)	74.1%	—
Remove log1p (22 feat + MSE + XGB)	70.4%	-3.7 pp
Remove features (3 raw + log1p + XGB)	42.9%	-31.2 pp
Remove both (3 raw + MSE + XGB, baseline)	58.2%	-15.9 pp

5.4 Gradient Surrogate Results

Scalar loss baseline. Before predicting gradients, scalar loss prediction validates that the parameter space is learnable (Table 5.9). XGBoost achieves Acc@10% above 96% on both gradient datasets. This is substantially easier than the forward block stress problem because the mesh (body 1) self stress is roughly the same size whether the mesh has buckled or not (a clean step at the bifurcation rather than a noisy valley).

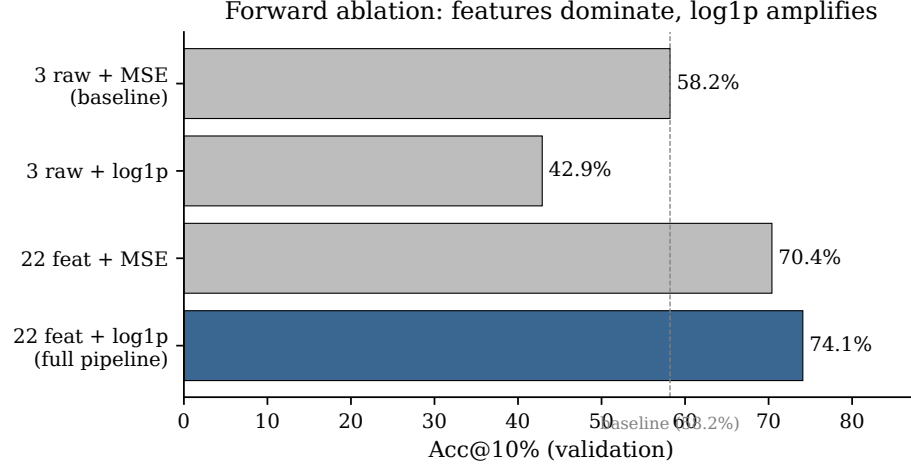


Figure 5.4: Forward surrogate ablation. Removing the 22 features informed by physics costs 31.2 pp; removing log1p alone costs only 3.7 pp. log1p without features is worse than the raw baseline because log compression amplifies confusion in transition samples until features separate them.

Table 5.9: Scalar loss prediction baseline (XGBoost, 22 features + log1p).

Dataset	R^2	MAPE	Acc@10%
Material (8,606)	0.989	1.5%	97.3%
Geometric (9,653)	0.989	2.5%	96.4%

Method progression. Table 5.10 shows the methodological progression for dL/dE . The autograd baseline (differentiating a scalar value MLP via PyTorch autograd) achieves only $\sim 30\%$ Acc@20%. Sobolev training⁵ adds gradient supervision via second order backpropagation and improves to 36.9% (+ ~ 7 pp). Switching to the decomposition into sign and the logarithm of magnitude with XGBoost as the learner jumps to 66.9% Acc@20% (+30 pp). The multihead neural network adds another 1.8 pp (68.7%) under proper 5-fold CV. The decomposition itself contributes the bulk of the gain across the model classes tested.

Table 5.10: dL/dE method progression. NN result is mean $\pm$ std over 5 folds.

Method	Sign%	$R^2_{\log}$	Acc@10%	Acc@20%
Autograd baseline	86.9%	0.753	14.1%	$\sim 30\%$
+ Sobolev supervision	89.3%	0.853	20.3%	36.9%
+ Sign+logmag (XGBoost)	95.2%	0.956	51.0%	66.9%
+ Multihead NN (5-fold CV)	97.2 $\pm$ 0.3%	0.967	51.5 $\pm$ 1.9%	68.7 $\pm$ 1.3%

Three gradients compared. The difficulty ordering across the three gradients

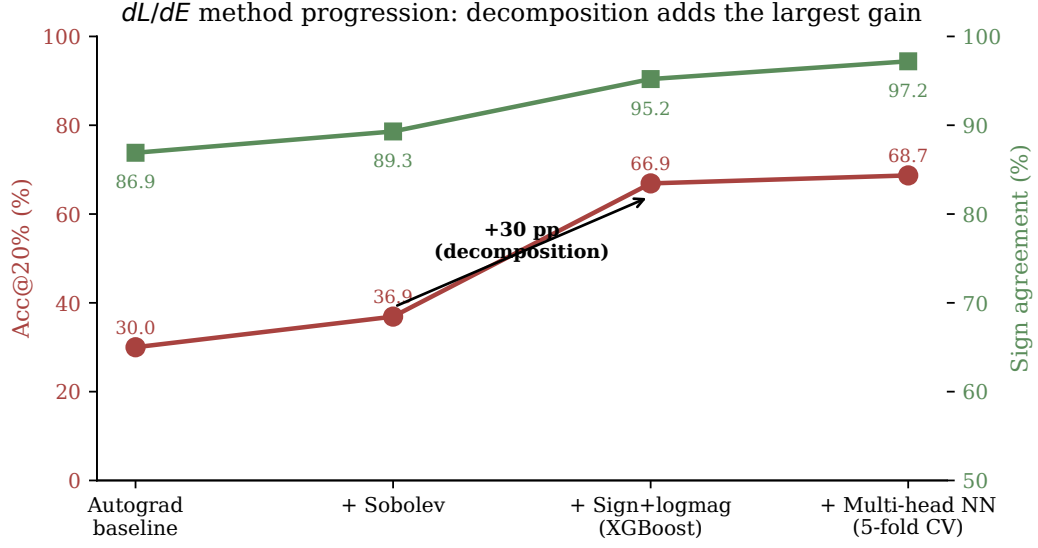


Figure 5.5: Method progression for dL/dE . Sobolev supervision adds ~ 7 pp over autograd; decomposing into sign and the logarithm of magnitude adds +30 pp on top, the largest single gain. Switching from XGBoost to a multihead neural network adds at most 1.8 pp further.

reflects physics. dL/dh is smoothest (beam bending scaling $\sigma \propto 1/h^2$), dL/dE has mixed regimes from material geometry interaction, and $dL/d\theta$ contains the bifurcation between buckling modes. Under the proper 5-fold CV protocol, the multihead NN and XGBoost are within ~ 1 pp on the smoother gradients (dL/dE : 68.7% vs. 66.9%; dL/dh : 74.5% vs. 75.8%), with XGBoost slightly ahead on dL/dh (Table 5.11). On $dL/d\theta$ XGBoost is markedly better than the NN (57.4% vs. $46.7 \pm 1.7\%$). The NN’s smooth, high capacity function approximator appears to overfit regional noise in the region after the mesh has buckled, noise that XGBoost’s axis aligned splits ignore. Both methods remain far below the ceiling of the smooth gradients on $dL/d\theta$, which is consistent with a difficulty imposed by the physics rather than a limitation of model capacity.

Regional decomposition for $dL/d\theta$. The drop in both Acc@20% and sign agreement once θ crosses the buckling boundary near 88° mirrors the noisy valley failure of the forward surrogate (Table 5.12). The performance in the smooth region ($\sim 50\%$ Acc@20%) is also notably below the headline dL/dh accuracy (74.5%). Even outside the bifurcation, $dL/d\theta$ is harder than dL/dh because the partial derivative with respect to a geometric angle couples both shape change and contact area change in a way that the simpler scalar parameters do not.

Table 5.11: Three gradient final results. All NN entries are proper 5-fold CV mean $\pm$ std with early stopping on an inner validation split; XGBoost rows are single train/test split.

Gradient	Method	Sign%	Acc@20%	Acc@10%	$R^2_{\log}$
dL/dh	Multihead NN (5-fold)	98.6 $\pm$ 0.3%	74.5 $\pm$ 1.0%	60.5 $\pm$ 2.6%	0.970
dL/dh	XGBoost decomp	98.5%	75.8%	65.4%	0.973
dL/dE	Multihead NN (5-fold)	97.2 $\pm$ 0.3%	68.7 $\pm$ 1.3%	51.5 $\pm$ 1.9%	0.967
dL/dE	XGBoost decomp	95.2%	66.9%	51.0%	0.956
$dL/d\theta$	Multihead NN (5-fold)	94.8 $\pm$ 0.3%	46.7 $\pm$ 1.7%	30.7 $\pm$ 1.7%	0.777
$dL/d\theta$	XGBoost decomp	94.9%	57.4%	37.2%	0.834

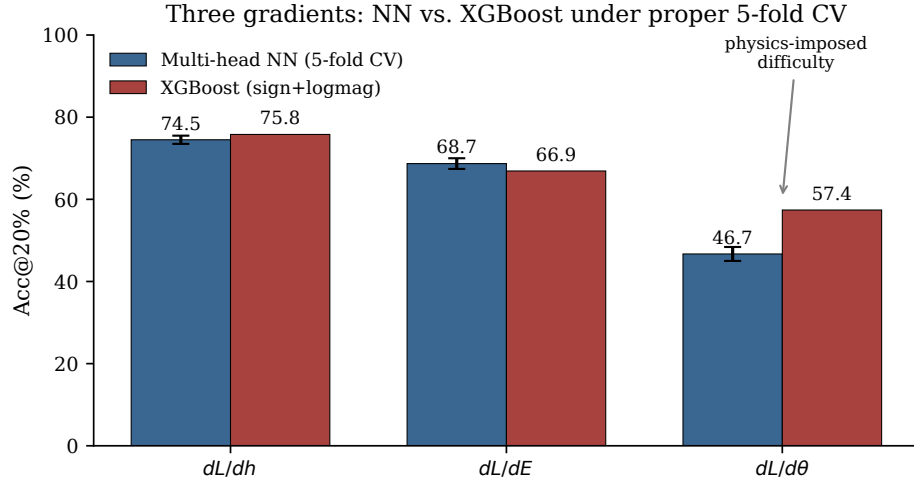


Figure 5.6: Three gradient final comparison. Under proper 5-fold CV the multi-head neural network and XGBoost are within ~ 1 pp on the two smoother gradients (dL/dE , dL/dh); on $dL/d\theta$ XGBoost is markedly stronger (57.4% vs. $46.7 \pm 1.7\%$), but both remain far below the ceilings on the smoother gradients, consistent with a difficulty imposed by the physics rather than a limitation of model capacity.

5.5 Cost and Speedup

The headline observation is a wide gap in per-query latency. A single FEM evaluation takes minutes of CPU time, while surrogate inference takes well under a millisecond on a laptop CPU. Table 5.13 summarises the orders of magnitude. The training cost of the surrogate is paid once and is small relative to the data generation cost, so the speedup applies to every downstream design query.

Table 5.12: $dL/d\theta$ regional decomposition. Out of fold predictions from the proper 5-fold CV multihead NN.

θ region	n	Sign%	Acc@20%	Acc@10%	Regime
$[60^\circ, 75^\circ)$	2,908	99.1%	53.0%	40.2%	Smooth
$[75^\circ, 85^\circ)$	1,915	97.3%	48.7%	35.3%	Smooth
$[85^\circ, 91^\circ)$	1,120	95.7%	49.2%	30.1%	Transition
$[91^\circ, 100^\circ)$	1,710	89.5%	42.7%	21.4%	Hardest
$[100^\circ, 111^\circ)$	2,000	90.0%	37.9%	20.8%	Hard

Table 5.13: Per query latency of FEM versus surrogate, by stage. Numbers are reported as orders of magnitude; exact values depend on the design point and hardware.

Stage	Forward (Exp. 1)	Gradient (Exp. 2)
FEM call (per sample, CPU)	minutes	minutes (longer than forward)
Surrogate training (one off)	seconds	tens of minutes (5-model ensemble)
Surrogate inference (per sample, CPU)	~ 0.03 ms	< 0.1 ms (3 gradients)
Inference speedup over FEM	$\sim 10^7 \times$	$\sim 10^7 \times$

5.6 Deployment Analysis

The accuracy numbers reported throughout this chapter range from nearly perfect ($\sim 99\%$) in smooth physical regimes to as low as $\sim 40\%$ in regions past the bifurcation. A single aggregate number would either oversell the smooth regions or unfairly penalise the surrogate’s headline accuracy with the difficulty of the hardest regions. The more useful question for a downstream user is: *which region and which workflow am I in, and is the surrogate accurate enough there?* This section answers that question by mapping the regional accuracy results back to representative use cases, and articulates a two tier protocol in which the surrogate and PolyFEM cooperate rather than compete.

Regional accuracy. Table 5.14 consolidates the headline numbers by physical regime. The forward surrogate column reports Acc@10% on the test set; the gradient columns report Acc@20% for the multihead NN under proper 5-fold cross validation. The rightmost column gives a qualitative confidence band that subsequent paragraphs use to discuss workflow fit. The dL/dh column is approximately constant across θ because h is a wall thickness parameter whose sensitivity is governed by bending mechanics that vary smoothly across the bifurcation. The other two columns track the difficulty induced by the physics. Forward accuracy and $dL/d\theta$ both dip in the buckling band, and the latter never recovers past $\sim 50\%$.

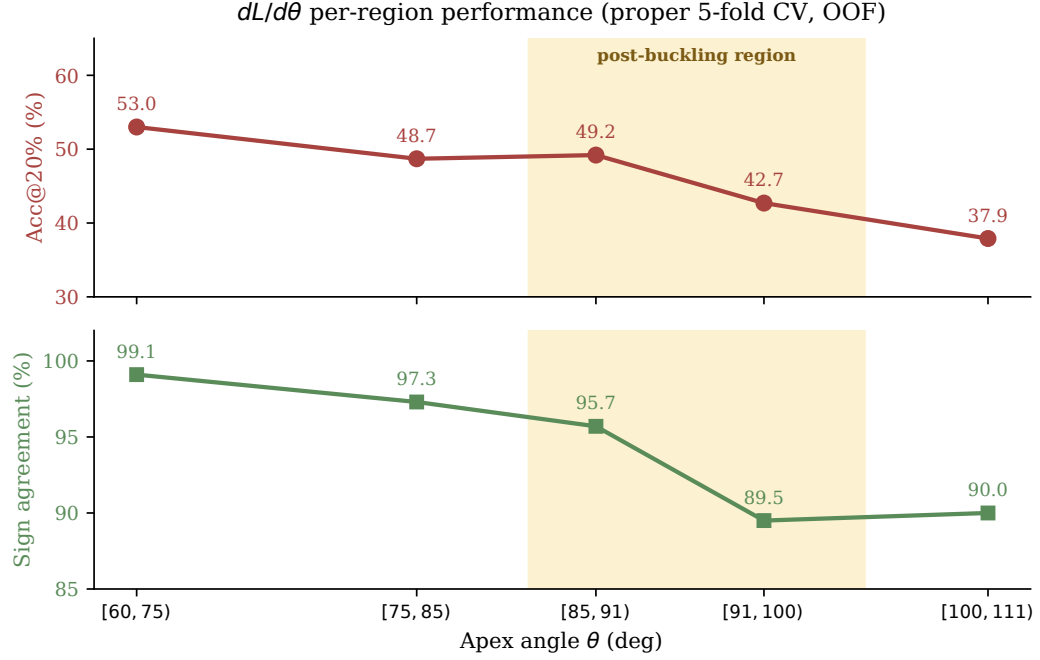


Figure 5.7: Regional performance by θ for $dL/d\theta$ from the proper 5-fold CV multihead NN (out of fold predictions, $n = 9,653$ in total). Smooth regions before the bifurcation reach $\sim 50\%$ Acc@20% with 97–99% sign agreement; the region past the bifurcation $[91^\circ, 100^\circ)$ drops to 42.7% Acc@20% and 89.5% sign agreement, with the high- θ region (which contains both stretching recovery effects and persistent post-buckling noise) the lowest at 37.9%.

High confidence: surrogate alone. For θ values outside the buckling band ($\theta < 85^\circ$ or $\theta \geq 100^\circ$, with the partial exception of $dL/d\theta$ at high θ), all three surrogate outputs are accurate enough that direct use is appropriate. This enables design exploration with many queries (a typical sensitivity sweep over the (θ, h, E) grid evaluates tens of thousands of design points, which is impractical at FEM’s minutes per call but completes in seconds with the surrogate, at $> 90\%$ Acc@10% in the smooth regimes), gradient based optimisation outer loops (the surrogate’s $> 70\%$ Acc@20% combined with $\geq 95\%$ sign agreement on dL/dh and dL/dE is sufficient for descent direction estimation under standard Armijo line search), and inverse design and search over latent space (Bayesian optimisation¹² or evolutionary search require thousands of forward evaluations per decision, and the surrogate is the only practical option there).

Mixed confidence: transition zone $[85^\circ, 95^\circ]$. The transition zone hosts the bifurcation between buckling modes and is the hardest region of the design space.

Table 5.14: Regional accuracy across the three surrogate outputs and the corresponding confidence band for deployment.

θ regime	n (fwd test)	Fwd Acc@10%	dL/dh Acc@20%	$dL/d\theta$ Acc@20%	Confidence band
$\theta < 80^\circ$	268	98.9%	$\sim 76\%$	53.0%	High
$[80^\circ, 87^\circ)$	186	83.3%	$\sim 76\%$	48.7%	High
$[87^\circ, 89^\circ)$	458	59.6%	$\sim 75\%$	$\sim 49\%$	Mixed
$[89^\circ, 93^\circ)$	742	69.3%	$\sim 75\%$	$\sim 49\%$	Mixed
$[93^\circ, 100^\circ)$	96	77.1%	$\sim 74\%$	42.7%	Low (verify)
$\theta \geq 100^\circ$	139	85.6%	$\sim 74\%$	37.9%	Low (verify)

Forward accuracy drops to $\sim 60\%$ Acc@10% at the worst point, and $dL/d\theta$ accuracy is roughly 42–49% Acc@20%. The surrogate is still useful here for coarse screening (even at 60% Acc@10%, the surrogate distinguishes designs with high stress $\sigma > 5$ MPa from designs with low stress $\sigma < 1$ MPa with high reliability, because those scales differ by far more than the surrogate’s relative error) and as an initial guess for FEM solvers (a gradient direction that is 50–60% accurate is still strictly better than a starting point with no information, and reduces the number of FEM Newton iterations needed at the final verification step). What the surrogate *cannot* do reliably in this regime is serve as the sole input to a safety critical decision or replace FEM for the single most important design candidate.

Low confidence: FEM verification. For the $dL/d\theta$ region past the bifurcation ($\theta > 91^\circ$), 46.7% Acc@20% places the surrogate below what a decision in structural engineering can tolerate without independent verification. The recommended workflow is the standard two tier protocol in which the surrogate and FEM cooperate: use the surrogate for the exploration phase with many queries (typically thousands to millions of design evaluations); identify the top- k candidate designs by surrogate score, e.g., $k = 100$ from a population of 10^6 ; reevaluate the top- k candidates with PolyFEM directly. The top- k verification step is orders of magnitude cheaper than running PolyFEM on the entire exploration population, and the final decision is now backed by the ground truth of the physics. This is the same multifidelity workflow recommended by Peherstorfer et al.²⁰ for uncertainty propagation, and is broadly used in surrogate assisted optimisation.

Decision rule. Table 5.15 consolidates the deployment recommendation into a short rule indexed by θ and use case. The future work outlined in Chapter 6 on conformal prediction and active learning aims to make these boundaries ones the surrogate itself can recognise, rather than ones the user has to keep track of manually.

Table 5.15: Deployment decision rule indexed by θ regime and target metric.

Condition	Recommended use
$\theta \in [60^\circ, 85^\circ]$ or many-query exploration	Use the surrogate directly.
$\theta \in [85^\circ, 95^\circ]$	Use the surrogate for coarse screening; verify any final design in this region with one PolyFEM call.
$\theta \in [91^\circ, 111^\circ]$ and target is $dL/d\theta$	Treat the surrogate as a ranking heuristic only; a FEM call is mandatory at the verification step.

5.7 Discussion

The two surrogates share a common pattern: a single algorithmic insight delivers the bulk of the gain, and the remaining headroom is set by physics rather than by the choice of model. For the forward surrogate the insight is feature engineering informed by physics (+15.9 pp); for the gradient surrogate it is the decomposition into sign and the logarithm of magnitude (+30 pp). In both cases tree models with the right inputs and targets reach within a few percentage points of much larger neural models, consistent with Grinsztajn et al.’s thesis¹⁰ on tabular ML, and a reminder that deep learning is not automatically the right tool for every regression problem.

The physical ceiling phenomenon is the more general takeaway. In Experiment 1, the 59.6% Acc@10% in the transition zone appears regardless of model architecture. It is the noisy valley of body 2’s response to force transmission, predicted by Gibson and Ashby’s cellular solids theory⁸ as the transition between stretching dominated and bending dominated deformation. In Experiment 2, both XGBoost and the multihead NN underperform on $dL/d\theta$ (XGBoost 57.4%, NN $46.7 \pm 1.7\%$ Acc@20%) compared with the $\sim 75\%$ they achieve on the smoother dL/dh . This gap is consistent with the FEM solver taking different convergence paths once the mesh has buckled rather than a model capacity limitation. These results suggest that surrogates for physics problems may be bounded by the shape of the underlying physical response, and that decomposing error by physical region, rather than reporting only aggregate metrics, is essential to tell modeling failure apart from a physical limit.

A high R^2 can mislead. The forward surrogate’s $R^2 = 0.97$ coexists with a 26% failure rate at the engineering tolerance.¹⁵ Metrics relevant for deployment (Acc@ $K\%$ for forward, sign agreement plus Acc@20% for gradient) reveal this mismatch and are

recommended as complementary reporting standards for surrogate work in physics informed ML.

Chapter 6

Conclusion

6.1 Contributions

This thesis develops two complementary machine learning surrogates over the design space (θ, h, E) of triangular mesh impact simulation, replacing FEM evaluation costs on the order of minutes per call with sub-millisecond inference. On test sets the models never saw during training:

- **Forward surrogate (block peak Von Mises stress):** XGBoost with 22 features informed by physics and a $\log(1 + y)$ target transform achieves $R^2 = 0.9684$, $\text{MAPE} = 10.2\%$, and $\text{Acc@10\%} = 74.1\%$ on 1,889 test samples, with inference under one millisecond per sample, roughly seven orders of magnitude faster than the FEM forward solve.
- **Gradient surrogate (smooth-max Von Mises sensitivities):** A multihead neural network with a decomposition into sign and the logarithm of magnitude reaches $\text{Acc@20\%} = 74.5 \pm 1.0\%$ and $\text{Sign} = 98.6 \pm 0.3\%$ for dL/dh , $\text{Acc@20\%} = 68.7 \pm 1.3\%$ and $\text{Sign} = 97.2 \pm 0.3\%$ for dL/dE , and $\text{Acc@20\%} = 46.7 \pm 1.7\%$ and $\text{Sign} = 94.8 \pm 0.3\%$ for $dL/d\theta$, all under proper 5-fold cross validation. Inference under 0.1 millisecond per sample, again about seven orders of magnitude over the FEM forward and adjoint pass.

Three findings emerge from the experiments. First, on the forward task, feature engineering grounded in the underlying physics accounts for the bulk of the gain. The 22-feature pipeline contributes +31.2pp over the raw input baseline (Section 5.3), and $\log_{1p}$ alone contributes only +3.7pp on top. Second, on the gradient task,

decomposing each target into its sign and the logarithm of its magnitude carries the bulk of the gain and works regardless of model family. XGBoost and the multihead neural network land within a few percentage points of each other across the three components (Section 5.4). Third, surrogate difficulty maps directly to the underlying physics. Smooth regions reach nearly perfect prediction, while the transition zone between buckling modes near $\theta \approx 88^\circ$ creates a noise floor that no model architecture overcomes.

Taken together, these findings define the overarching contribution: extending FEM surrogate modeling from the smooth regime it usually assumes into a regime with a structural discontinuity. The existing surrogate literature reports high aggregate R^2 on smooth parameter-to-response maps; this thesis shows what it takes to build a usable surrogate when a buckling bifurcation makes the forward response a noisy valley and makes the gradients flip sign across many orders of magnitude. The three ingredients that make it work (features informed by physics, the sign and log-magnitude decomposition, and deployment relevant metrics that expose where aggregate R^2 is misleading) are the practical answer to that harder problem, and each is shown to matter by ablation.

6.2 Limitations

- **Transition zone reliability.** For $\theta \in [85^\circ, 95^\circ]$, the forward surrogate drops to 59.6% Acc@10% and the gradient surrogate to 42.7% Acc@20% on $dL/d\theta$. Safety critical decisions in this region require FEM verification.
- **No extrapolation.** The surrogate is reliable only within training parameter bounds and only for θ values within the training grid. Leave- θ -group-out cross validation drops Acc@10% from 73.4% to 50.4%, suggesting heavy reliance on interpolation between dense neighboring training points.
- **No uncertainty quantification.** Both surrogates output point estimates. Ensemble variance from the multihead NN can serve as an uncalibrated proxy but is not formally validated.
- **Single loading condition and material model.** All simulations use Neo-Hookean elasticity²¹ with frictionless contact and a single impact scenario. Plasticity, damage, dependence on rate, friction, and oblique or repeated loading

would require dataset regeneration and likely retraining.

- **Physical ceiling on the noisy valley.** The 74.1% Acc@10% on the forward task and the 42.7% Acc@20% on $dL/d\theta$ in the region past the bifurcation may reflect noise that no model architecture or amount of training data can remove in the underlying simulation (mesh sensitivity, contact resolution, Newton convergence path), in which case no surrogate can exceed this ceiling regardless of architecture or data scale.

6.3 Future Work

- **Integration with PolyFEM.** The work in this thesis treats PolyFEM as an offline data source. The FEM solver writes simulation outputs to disk, the surrogates are trained separately, and inference is then performed by standalone Python code. The next step is to expose the trained surrogates as a stable callable interface inside the PolyFEM ecosystem. PolyFEM’s Python bindings (`polyfem-python`) could then swap the surrogate in for the FEM forward and adjoint passes whenever the user accepts approximate evaluation. The accompanying code repository already organises the surrogates as an importable Python package (`polyfem_ml`) with two top level classes, `StressSurrogate` and `GradientSurrogate`, each providing a single `.predict(theta, h, E)` entry point that hides the feature pipeline, target transform, normalisation, and decomposition decoder. This is an intentionally narrow API surface designed to slot into a future PolyFEM configuration flag along the lines of `solver = "surrogate"`, with FEM verification triggered only when the surrogate’s predicted error band crosses a decision boundary.
- **Joint 7-head model.** The current gradient surrogate trains a 3-head model for dL/dE and a 5-head model for $[dL/dh, dL/d\theta]$ separately. A unified 7-head model with a single shared backbone may exploit cross task structure to improve $dL/d\theta$ accuracy.
- **Conformal prediction intervals.** Post hoc conformal calibration²³ would attach distribution free coverage guarantees to surrogate outputs, enabling principled fallback to FEM when the predicted interval crosses a decision boundary.

- **Active learning in the transition zone.** Using Expected Improvement criteria¹² to direct additional FEM evaluations to $\theta \in [85^\circ, 95^\circ]$ could narrow the gap to the physical ceiling with the smallest possible compute investment.
- **Multifidelity extension.** Coarse mesh FEM as a low fidelity gradient source,²⁰ with the surrogate learning a correction to the high fidelity adjoint, may improve transition zone accuracy.
- **End to end integration into the optimization loop.** The proof of concept demonstration is to embed the surrogate as a drop in replacement for FEM backward passes inside a gradient based design optimizer, comparing convergence speed and quality of the final design against optimization with direct FEM.

Appendix A

Reproducibility Details

A.1 Software and Versions

All FEM simulations use PolyFEM²² with its Python interface. ML training uses scikit-learn for Random Forest, GBR, and SVR; XGBoost for tree boosting; PyTorch for the multihead neural network. The forward surrogate uses a fixed random seed of 42 for all stochastic components (data split, XGBoost, RF, GBR, MLP, SVR). The multihead NN gradient surrogate uses outer cross validation seed 42. Per-fold model seeds are $42 + i$ for $i = 0, \dots, 4$.

A.2 Hyperparameter Summary

XGBoost (forward): 5,000 trees, `max_depth` = 8, `learning_rate` = 0.01, `subsample` = 0.8, `colsample_bytree` = 0.7, $\gamma = 0$, $\lambda = 1$, target $\log(1 + y)$ with inverse $\exp(\hat{z}) - 1$ at decode.

Random Forest (forward): 3,000 trees, fully grown (`min_samples_leaf` = 1), `max_features` = 0.5.

XGBoost (gradient): 500 trees, `max_depth` = 8, `learning_rate` = 0.05, `colsample_bytree` = 0.8. Sign head uses `binary:logistic` objective. Magnitude head uses `reg:squarederror` on $\log_{10}(|g| + 1)$. Decode: $\hat{g} = \text{sgn}(\hat{s}) \cdot (10^{\hat{m}} - 1)$.

Multihead NN: Shared backbone 6 Linear+SiLU stages, hidden width 512. Each head is Linear-SiLU-Linear with hidden width 512. AdamW, `lr` = 5×10^{-4} ,

weight_decay = 10^{-5} . Cosine annealing to $T_{\max} = 400$. Batch size 512. Gradient clipping max_norm = 1. Early stopping on the inner validation fold, monitoring Acc@20%, evaluated every 20 epochs, with patience 40.

Loss ramp: $\lambda_{\text{ramp}}(\text{epoch}) = \min(1, \max(0, (\text{epoch} - 30)/40))$. Warmup on the value loss alone for the first 30 epochs; the full weight on the gradient loss is reached at epoch 70.

Gradient feature set: The XGBoost gradient baseline uses a 15-dim feature set. This is the three raw inputs plus the same transforms informed by physics as the forward set, but *without* the seven-level h one-hot, since the gradient datasets sample h on a different (5-level) grid.

A.3 Feature List

The 22 features informed by physics for the forward surrogate are: the three raw inputs θ , h , E ; the log transforms $\log h$, $\log E$; the trigonometric encoding $\sin(\theta\pi/180)$; features encoding the distance to the transition $|\theta - 88|$, $(\theta - 88)^2$, $(\theta - 88)^3$; bending stiffness proxies h^2E , h^3E ; compliance h/E ; pairwise interactions θh , θE , hE ; and the seven-level h one-hot encoding (one indicator per discrete thickness value).

A.4 Dataset Provenance

The 18,887 forward simulations and the 8,606 material gradient and 9,653 geometric gradient samples were generated on Compute Canada CPU resources between April and May 2026. Each FEM run logs the full transient stress trajectory; the forward dataset records the temporal maximum on body 2; the gradient datasets record the discrete adjoint gradients of the smooth-max Von Mises objective on body 1 at the final converged time step.

A.5 Code and Data Availability

The source code, trained checkpoints, and reproduction instructions live in a companion repository:

<https://github.com/jczhao1/polyfem-ml-surrogate>

The repository is organised as a single Python package (`polyfem_ml`) plus per-experiment training scripts.

Inference (the recommended entry point)

After installing the package with `uv sync` or `pip install -e .`, the published checkpoints can be loaded with three lines:

```
from polyfem_ml import StressSurrogate, GradientSurrogate

stress = StressSurrogate.load_default()
sigma_peak = stress.predict(theta=87.3, h=0.05, E=18.0)

grad = GradientSurrogate.load_default()
dE, dh, dtheta = grad.predict(theta=87.3, h=0.05, E=18.0)
```

Both `.predict()` methods accept scalar or 1-D NumPy array inputs and return predictions in under one millisecond on CPU. Trained checkpoints ship inside `polyfem_ml/checkpoints/`.

Retraining from scratch

The repository contains the full training pipeline. With the raw PolyFEM data bundles in place (see `data/README.md`):

```
# Forward surrogate (~2 min)
uv run python experiments/01_forward/00_split_data.py
uv run python experiments/01_forward/01_train_eval.py

# Gradient surrogate (~15 min each, can run in parallel)
OMP_NUM_THREADS=1 uv run python experiments/02_gradient/00_prepare_data.py
OMP_NUM_THREADS=1 uv run python experiments/02_gradient/01_proper_cv_dLdE.py
OMP_NUM_THREADS=1 uv run python experiments/02_gradient/02_proper_cv_htheta.py
```

Each training script publishes its best-fold model and normalisation statistics back into `polyfem_ml/checkpoints/`, so subsequent inference calls automatically pick up the retrained model. `OMP_NUM_THREADS=1` is required on macOS to avoid threading deadlocks between PyTorch and the BLAS used by NumPy and scikit-learn.

Repository layout

`polyfem_ml/` Python package: data loading, 22- and 15-dim feature builders, metrics, the `MultiHead3 / MultiHead5` torch modules, shared training utilities, the public `inference.py` API, and the checkpoint directory.

`experiments/01_forward/` Forward surrogate training scripts and JSON result files.

`experiments/02_gradient/` Gradient surrogate training scripts (proper 5-fold CV + XGBoost baseline) and JSON result files.

`tools/check_inference.py` End to end smoke test that loads checkpoints and runs predictions on representative design points.

`pyproject.toml` *and* `uv.lock` Pinned dependencies for fully reproducible installs.

Bibliography

- ¹ Kamyar Azizzadenesheli, Nikola Kovachki, Zongyi Li, Miguel Liu-Schiaffini, Jean Kossaifi, and Anima Anandkumar. Neural operators for accelerating scientific simulations and design. *Nature Reviews Physics*, 6(5):320–328, 2024.
- ² Pranav Vijay Badarinath and Maria Chierichetti. A machine learning approach as a surrogate for a finite element analysis: Status of research and application to one dimensional systems. *Sensors*, 21(5):1654, 2021.
- ³ Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.
- ⁴ Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 785–794, 2016.
- ⁵ Wojciech M. Czarnecki, Simon Osindero, Max Jaderberg, Grzegorz Świrszcz, and Razvan Pascanu. Sobolev training for neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- ⁶ Vikram S. Deshpande, Norman A. Fleck, and Michael F. Ashby. Effective properties of the octet-truss lattice material. *Journal of the Mechanics and Physics of Solids*, 49(8):1747–1769, 2001.
- ⁷ Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232, 2001.
- ⁸ Lorna J. Gibson and Michael F. Ashby. *Cellular Solids: Structure and Properties*. Cambridge University Press, 2nd edition, 1997.
- ⁹ Alesandro E. Gongora, Caleb Friedman, Deryck K. Newton, Timothy D. Yee, Zachary Doorenbos, Brian Giera, Eric B. Duoss, Thomas Y.-J. Han, Kyle Sulli-

- van, and Jennifer N. Rodriguez. Accelerating the design of lattice structures using machine learning. *Scientific Reports*, 14:13703, 2024.
- ¹⁰ Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems (Datasets and Benchmarks)*, 2022.
- ¹¹ Junyan He, Seid Koric, Diab Abueidda, Ali Najafi, and Iwona Jasiuk. Sequential deep operator networks (S-DeepONet) for predicting full-field solutions under time-dependent loads. *Engineering Applications of Artificial Intelligence*, 127:107258, 2024.
- ¹² Donald R. Jones, Matthias Schonlau, and William J. Welch. Efficient global optimization of expensive black-box functions. *Journal of Global Optimization*, 13:455–492, 1998.
- ¹³ Ron Kohavi. A study of cross-validation and bootstrap for accuracy estimation and model selection. In *Proceedings of the 14th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 1137–1145, 1995.
- ¹⁴ Nikola Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023.
- ¹⁵ Jakub Kudela and Radomil Matousek. Recent advances and applications of surrogate models for finite element method computations: A review. *Soft Computing*, 26(24):13709–13733, 2022.
- ¹⁶ Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations (ICLR)*, 2021.
- ¹⁷ Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021.

- ¹⁸ Lu Lu, Xuhui Meng, Shengze Cai, Zhiping Mao, Somdatta Goswami, Zhongqiang Zhang, and George Em Karniadakis. A comprehensive and fair comparison of two neural operators (with practical extensions) based on FAIR data. *Computer Methods in Applied Mechanics and Engineering*, 393:114778, 2022.
- ¹⁹ Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- ²⁰ Benjamin Peherstorfer, Karen Willcox, and Max Gunzburger. Survey of multi-fidelity methods in uncertainty propagation, inference, and optimization. *SIAM Review*, 60(3):550–591, 2018.
- ²¹ Ronald S. Rivlin. Large elastic deformations of isotropic materials. IV. further developments of the general theory. *Philosophical Transactions of the Royal Society A*, 241(835):379–397, 1948.
- ²² Teseo Schneider, Jérémie Dumas, Xifeng Gao, Mario Botsch, Daniele Panozzo, and Denis Zorin. Poly-spline finite-element method. *ACM Transactions on Graphics*, 38(3), 2019.
- ²³ Glenn Shafer and Vladimir Vovk. A tutorial on conformal prediction. *Journal of Machine Learning Research*, 9:371–421, 2008.
- ²⁴ Richard von Mises. Mechanik der festen körper im plastisch-deformablen zustand. *Nachrichten der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, pages 582–592, 1913.
- ²⁵ Jeffrey M. Wooldridge. *Introductory Econometrics: A Modern Approach*. Cengage Learning, 7th edition, 2019.